Xingbo Fu

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EDUCATION AND EXPERIENCE

Xi'an Jiaotong University, Xi'an, China

Master of Science, Control Science and Engineering 2017.09-2020.06

Columbia University, New York, NY, USA

Research assistant, Data Science Institute 2019.06~2019.09

Xi'an Jiaotong University, Xi'an, China

Bachelor of Science, Automation 2013.09-2017.06

RESEARCH PROJECTS

Driver Behavior Modeling with Deep Reinforcement Learning 2019~

- Modeled human-like driving behavior in car-following problems using deep reinforcement learning.
- Implemented a deep deterministic policy gradient (DDPG) model and proposed the Twin Delayed Deep Deterministic policy gradient (TD3) algorithm with attention mechanism to determine driver behavior.
- Evaluated performance against other seven baseline algorithms.

Analyzing Spatiotemporal Data with Deep Learning 2018~2019

- Proposed a framework of spatiotemporal attention networks (STAN).
- Employed a multi-head self-attention mechanism to extract spatial correlations and captured temporal dependencies by Seq2Seq with a global attention mechanism.
- Compared performance with seven baseline approaches.

Big Data Computing Platform for Energy Internet 2017~2019

- Built Hadoop-Spark big data computing clusters for data processing and storage.
- Used Multithread Technology and TCP socket to collect data and exchange messages with computing clusters.
- Received data by Spark Streaming and processed data by Spark MLlib.
- Utilized Spark SQL to store data in distributed systems (Hive).
- Obtained a software copyright of big data computing platform.

Solar Irradiance Data Simulation 2017~2019

- Proposed a solar irradiance simulation method based on feature clustering and Markov transition probability matrix.
- Accomplished this method with solar irradiance data for one station according to state reduction.
- Generalized this method to multi-station solar irradiance data.

TECHNICAL AND OTHER SKILLS

Core Domain Expertise

Machine learning algorithm, Big data technology, Deep learning

Programming and Software

Python, Java, Scala, Hadoop, Spark, MySQL

PUBLICATIONS

Preprint

1. **When Do Drivers Concentrate? Attention-based Driver Behavior Modeling With Deep Reinforcement Learning [PDF]**
   ***Fu Xingbo**, Di Xuan, Mo Zhaobin*

Conference

2. **Spatiotemporal Attention Networks for Wind Power Forecasting [PDF]**
   ***Fu Xingbo**, Gao Feng, Wu Jiang, Wei Xinyu, Duan Fangwei*
   International Conference on Data Mining Workshops (ICDM 2019 Workshops), Beijing, China.
3. **A Simulation Approach to Multi-Station Solar Irradiance Data Considering Temporal Correlations [PDF]**
   ***Fu Xingbo**, Gao Feng, Wu Jiang, Ruanming Huang, Yichao Huang, Fei Fei*
   International Conference on Innovative Smart Grid Technologies (ISGT Asia 2019), Chengdu, China.
4. **Wind Power Capacity Planning in Enterprise's Microgrid based on Approximation Expectation of Operation Cost [PDF]**
   *Zhou Yuzhou, Zhai Qiaozhu, **Fu Xingbo**, Guan Xiaohong, et al.*
   IEEE Power & Energy Society General Meeting (PESGM 2019), Atlanta, GA, USA.
5. **A Simulation Method of Solar Irradiance Data Based on Feature Clustering and Markov Transition Probability Matrix [PDF]**
   ***Fu Xingbo**, Gao Feng, Wu Jiang, Guan Xiaohong, et al.*
   World Congress on Intelligent Control and Automation (WCICA 2018), Changsha, China.
6. **Hybrid Features based K-means Clustering Algorithm for use in Electricity Customer Load Pattern Analysis [PDF]**
   *Pengyuan Liu, Chenye Yang, Jiang Wu, **Fu Xingbo**, Ruanming Huang, et al.*
   Chinese Control Conference (CCC 2018), Wuhan, China.

Patent

Gao F, Fu X, Wu J, Guan X, et al. A Novel Approach to Simulate Solar Radiation Data[J]. 2019. (Substantive examination)

Software Copyright

Big data platform for Energy Internet. 2018SR842914

SELECTED HONORS

Outstanding Graduate Student	2019
Outstanding Graduate	2017
Siyuan Scholarship	2016
Outstanding Student Leader	2016
Siyuan Scholarship	2015
Outstanding Student	2015
Siyuan Scholarship	2014
Outstanding Student Leader	2014

PERSONAL WEBSITE

http://www.xbfu.top